

User-Owned Preference Policies for Cross-Model, Domain-Conditioned Language-Model Personalization

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Abstract

Language-model products commonly keep response preferences inside one provider or conflate preferences with conversational memory. We formulate personalization as a user-owned behavioral policy: typed preferences are selected by domain and task, compiled into a provider-neutral instruction, and remain inspectable, reversible, and subordinate to the current request. We contribute an auditable representation with scope, confidence, provenance, and locks; five comparison conditions; a controlled multidomain benchmark; and a reproducible execution and blinded-evaluation pipeline. In an exploratory two-model local pilot comprising 128 real generations, a compact structured policy used 19.0 fewer input tokens per prompt than a short history-retrieval baseline, but produced 70.8 more words and added 3.12 seconds of latency on average. Its post-hoc reference-concept coverage was indistinguishable within a wide exploratory interval ($\Delta = 0.005$, 95% cluster-bootstrap interval $[-0.141, 0.177]$). The result shows that input-context compression alone does not establish end-to-end efficiency. No human ratings have yet been collected, so improved subjective quality remains an open hypothesis rather than a reported result.

1 Introduction

People use multiple large language models (LLMs), but each interface typically receives only a local account setting, a manually repeated instruction, or a provider-specific memory. The resulting problem is not only remembering what a person said. It is representing *how that person wants a response to be constructed for the task at hand* and transferring that representation without assuming that expertise or style in one domain applies everywhere.

We call this proposed layer Preference Intelligence. Its design goals are portability across model providers, contextual selection, and transparent user control. A response preference is an instruction-like behavioral constraint—such as desired depth, format, or use of examples—rather than a demographic label or an assertion about identity. The current request always outranks a stored default.

The primary research question is whether a dynamically learned, domain-conditioned, provider-independent preference representation improves subjective response quality across heterogeneous model families. We further ask whether it improves on static profiles and history retrieval, whether learning is reliable under drift and adversarial evidence, and whether structured profiles reduce context overhead.

The contributions of the present artifact are:

1. a formal, inspectable *preference policy* representation and applicability problem that separates response behavior from conversation memory;
2. a behavioral-conformance framing of portability: an unchanged profile must induce the intended scoped behavior across model families, not merely serialize successfully;

3. a benchmark and human-study protocol targeting domain isolation, preference learning, context efficiency, drift, and incorrect inference; and
4. a local-first reference implementation and reproducible comparison of five personalization conditions, including 128 real local-model generations and an observed prompt-compression/output-expansion trade-off.

The representation and pipeline are implemented contributions; the pilot is exploratory systems evidence. It does not establish the primary subjective-quality hypothesis. Irrelevant, stale, or incorrect preferences can still reduce quality or undermine user control.

2 Related Work

The relevant literature spans personalized language generation, user modeling, implicit feedback, adaptive interfaces, recommender systems, conversational memory, persona conditioning, and preference optimization. Context-aware recommender systems and implicit-feedback methods established that behavior is not equivalent to positive preference and that context belongs in the utility model [Hu et al., 2008, Adomavicius and Tuzhilin, 2011].

Recent LLM personalization work has moved beyond fixed persona sentences. LaMP established retrieval-based personalization tasks [Salemi et al., 2024]; other systems use per-user adapters or embeddings [Tan et al., 2024, Liu et al., 2025], domain-aware compressed profiles [Su et al., 2025], and latent preference attributes [Li et al., 2025]. PLUS jointly learns an interpretable natural-language summary and a personalized reward model [Nam et al., 2026], while S2Pref distinguishes stable from situational preferences and includes preference conflicts [Gao et al., 2026]. These results mean that dynamic profiles, domain conditioning, and interpretable summaries are not individually novel.

Evaluation work also warns against assuming that personalization helps. PrefEval tests explicit and implicit preference following in long contexts [Zhao et al., 2025], and PrefDisco reports settings where uninformed personalization underperforms a generic response [Li et al., 2026]. Broad memory systems offer temporal retrieval and compression mechanisms [Park et al., 2023, Packer et al., 2024, Chhikara et al., 2025] but usually optimize episodic or factual recall rather than an editable response-preference policy.

The narrower gap investigated here is the joint system of typed response dimensions, hierarchical applicability, calibrated evidence provenance, user governance, provider-neutral compilation, and behavioral cross-model evaluation. The structured literature review found no peer-reviewed demonstration of the entire bundle at its evidence cutoff; this is an absence-of-evidence statement, not a priority claim. The review protocol and study selection limitations are reported in the accompanying artifact.

3 Problem Formulation

Let u denote a user, x a current request, d its domain, t its task type, m a model provider/snapshot, and P_u the portable preference profile. A selector $S(P_u, d, t, x)$ returns a bounded set of relevant preference records, while compiler C maps those records to an instruction context c . Generation is

$$y \sim p_m(y \mid x, C(S(P_u, d, t, x))) . \quad (1)$$

The current request is a hard precedence constraint: if x conflicts with a stored default, x wins. Selection is therefore not merely nearest-neighbor retrieval; it combines scope compatibility, evidence authority, confidence, recency, locks, and conflict resolution.

The selector distinguishes two latent decisions. Evidence confidence estimates support that a stored assertion reflects the user’s preference; contextual applicability estimates whether that assertion should influence this request. A failure of the first is a *preference-inference error*; a failure of the second is a *preference-applicability error*. A user correction to the latter should not necessarily delete or globally weaken a correct preference.

For subjective utility $Q(u, x, y)$, the primary estimand is the average within-user contrast between domain-dynamic personalization and no personalization over a prespecified population of users, tasks, and model families. Portability additionally requires that P_u is unchanged across m ; thin message-format adapters do not count as model-specific preference storage. We therefore distinguish *syntactic portability* (the profile can be parsed and compiled for a new model) from *behavioral conformance* (the resulting response satisfies the same applicable policy without increasing correctness, safety, privacy, or cost failures beyond a prespecified margin).

The design treats correctness, safety, privacy exposure, context cost, and preference adherence as distinct outcomes. A system that increases stylistic adherence while decreasing correctness does not unconditionally improve quality.

4 Preference Representation

Each preference record contains a stable identifier, bounded dimension and value, scope tuple (d, s, t) for domain, subdomain, and task, confidence, lifecycle state, source type, evidence counts, timestamps, decay policy, user lock, user-authored evidence, and provenance. Null scope components form a global default; more specific compatible records override broader records for the same dimension.

This hybrid symbolic representation is human-readable and deterministically filterable while allowing learned components to propose updates. Natural-language compilation occurs only after selection. Assistant responses and webpage text are untrusted inputs and cannot independently become user evidence. Users can inspect, edit, lock, suppress, or delete values.

Not measured in the current study: Confidence calibration and observed editability require approved participant data. Schema conformance is covered by software tests, but is not a user outcome.

5 Dynamic Preference Learning

The proposed updater aggregates signed, timestamped evidence by dimension and scope. Explicit dashboard changes and direct statements have greater authority than repeated implicit corrections; one ambiguous event cannot create a durable high-confidence preference. A newer direct durable statement can supersede an unlocked inferred value, while weaker contradictory evidence maintains an explicit ambiguous state and causes selection to abstain. Locked values do not change automatically, but the conflict remains inspectable.

Recency decay is applied to inferential evidence, while user-locked preferences do not decay. Temporary, task-local requests affect the current generation but are not promoted to durable profile values without an explicit persistence cue. Drift is evaluated through stable change points, temporary overrides, and alternating task-dependent preferences.

The reference interface also accepts negative applicability evidence: “this preference was not relevant here” adds a scoped exception while retaining the underlying preference assertion. This creates direct supervision for the second error type without conflating it with preference truth.

Not measured in the current study: Learning curves, activation precision/recall, calibration, and adaptation-delay estimates require the frozen longitudinal experiments.

6 Domain-Conditioned Personalization

Domain conditioning prevents a local observation—for example, advanced Java knowledge—from becoming a general assumption about finance or physics. The selector admits global defaults and compatible domain, subdomain, and task records, resolves each dimension by specificity and authority, and applies a confidence threshold. Missing domain knowledge remains unknown.

The reference implementation operationalizes a separate applicability gate as a weighted combination of symbolic scope match, dimension–task relevance, and decayed evidence confidence. Scope mismatch is a hard rejection; locked state does not bypass it. The present coefficients are inspectable engineering priors, not fitted probabilities or experimental findings. Every application and rejection produces a bounded explanation trace for error analysis.

The controlled benchmark deliberately pairs high familiarity in one domain with low familiarity in another. It also includes ambiguous domains and explicit current-request overrides. The central mechanism ablation removes scope while holding the evidence and token budget fixed.

Not measured in the current study: Domain leakage, relevant-preference recall, and domain-conditioned versus global contrasts require the locked generation and blinded rating study.

7 Experimental Setup

We predefine five conditions: no personalization (A), a participant-authored static global profile (B), retrieval of relevant user-authored history (C), a dynamic global structured profile (D), and a dynamic domain-conditioned profile (E). Generation settings and neutral task instructions are held fixed within a comparison. Equal personalization-token budgets support the primary quality comparison; natural budgets support the efficiency comparison.

The synthetic benchmark contains multidomain personas, development and locked task-template families, explicit conflicts, invalid evidence, and preference drift. The proposed human evaluation is a randomized, blinded, repeated-measures incomplete block with participant and task effects. The primary outcome is pairwise E-versus-A preference with ties retained. Secondary outcomes include usefulness, clarity, depth/detail appropriateness, adherence, context tokens, latency, domain leakage, and incorrect activation.

Reproducibility. Every generation record includes the condition, provider and returned model identifiers, parameters, full prompts, selected preference records, compiled context, token usage or labeled approximation, latency, response, automatic diagnostics, error, timestamp, and seed. Run manifests contain input hashes and the code revision when available.

Executed local-model pilot. We instantiated one synthetic persona with advanced software-engineering and beginner-finance preferences and selected four tasks: Java virtual threads, Kafka rebalancing, bond duration, and portfolio diversification. Qwen 3 8B Q4_K_M and Llama 3.2 1B Q8_0 were served through a local Ollama endpoint. We disabled hidden reasoning to reserve a comparable 260-token visible-output budget, set temperature to 0.3, generated two stochastic replicates per cell, and deterministically shuffled cells using recorded seeds. The first run crossed both models with all five conditions (80 generations; seed 20260913). After observing excess structured-prompt overhead, we retained that run and executed a labeled post-hoc compiler ablation over A, C, and E (48 generations; seed 20260914). No failed cells were discarded.

Pilot analysis. A narrow reference-concept coverage diagnostic was derived after generation from factual reference points already present in the benchmark. It can miss valid paraphrases and is not a factuality or preference metric. We report paired differences and exploratory percentile intervals from 5,000 cluster-bootstrap draws over eight model-task clusters. Because the metric and compact ablation were post-hoc, we report no confirmatory p -values.

Human-study status. No participants have been recruited and no human ratings are available. Exact hosted-provider snapshots, participant count, and smallest effect of interest must be frozen in a preregistration and approved study protocol before confirmatory collection.

8 Exploratory Local-Model Results

All 128 assigned local-model generations completed without a transport failure. The primary five-condition run contributed 80 outputs and the compact-compiler ablation contributed 48. Raw responses, complete prompts, selected preference records, provider token counts, finish reasons, latency, run manifests, and seeds are retained in JSONL. The two analysis datasets also provide 64 and 32 condition-blinded response pairs, respectively, with mapping keys stored separately.

8.1 Five-condition pilot

The initial structured compiler (`verbose_v1`) did not deliver the hoped-for context advantage. Compared with history retrieval, E consumed 65.0 additional input tokens per prompt (exploratory 95% interval [62.1, 68.0]), produced 52.1 additional words ([11.6, 99.1]), and added 3.56 seconds of latency ([1.16, 6.26]). Its reference-concept coverage difference was -0.047 ($[-0.177, 0.104]$). Compared with no personalization, E produced 34.8 fewer words ($[-75.2, -7.0]$) and returned 1.13 seconds faster ($[-2.29, -0.13]$), but required 116.0 more input tokens. The coverage difference was 0.073 ($[-0.052, 0.188]$). These intervals describe this small post-hoc pilot and are not confirmatory confidence intervals for a target population.

8.2 Compact compiler ablation

We replaced descriptive structured records with fixed, bounded behavioral clauses (`compact_v2`), reducing the example compiled profile from 447 to 158 characters. The resulting E prompts used 19.0 fewer input tokens than history retrieval ($[-19.75, -18.25]$), thereby meeting a narrow input-context objective. However, E produced 70.8 more words ($[28.6, 117.3]$), 99.1 more output tokens ($[42.0, 161.8]$), and added 3.12 seconds of latency ($[1.32, 5.03]$). The rate of length-limited termination was 0.313 higher, although its interval included zero ($[-0.063, 0.750]$). Reference-concept coverage was nearly unchanged ($\Delta = 0.005$, $[-0.141, 0.177]$).

Table 1: Pooled paired E-minus-comparator differences across two models, four tasks, and two replicates. Intervals are exploratory model-task cluster bootstraps. Lower is better for token, word, and latency costs; reference coverage is a post-hoc lexical proxy.

Compiler	Comparator	Δ coverage	Δ input tokens	Δ words
verbose_v1	History retrieval	-0.047	+65.0	+52.1
verbose_v1	No personalization	+0.073	+116.0	-34.8
compact_v2	History retrieval	+0.005	-19.0	+70.8
compact_v2	No personalization	-0.021	+32.0	-16.7

Against no personalization, compact E used 32.0 additional input tokens ($[31.25, 32.75]$), produced 16.7 fewer words ($[-42.8, 6.3]$), and returned 1.47 seconds faster ($[-2.73, -0.47]$). Its coverage difference was -0.021 ($[-0.156, 0.125]$).

The strongest supported statement is therefore a systems trade-off: compact structured selection reduced input context relative to this history baseline, but the tested instruction semantics expanded generation enough to erase the end-to-end efficiency benefit. Neither run establishes preference satisfaction or subjective quality.

9 Ablation Studies

The broader planned ablations independently remove domain scope, confidence gating, recency decay, evidence-source weighting, provenance filtering, conflict handling, and user locks. Representation baselines compare structured records with a natural-language profile and multiple history retrievers. Tests hold evidence, task allocation, generation settings, and applicable token budget fixed.

The first completed ablation changed only the compiler surface form from a verbose record serialization to compact behavioral clauses. It reduced the representative compiled profile from 447 to 158 characters and reversed the input-token contrast against history retrieval from +65.0 to -19.0 tokens per prompt. It did not improve the exploratory reference proxy and increased both generated length and latency (Section 8). This identifies compiler wording and total generated tokens—not profile storage alone—as necessary experimental factors.

Not measured in the current study: The remaining preregistered component ablations and their full-versus-ablation effects with corrected intervals, including adherence precision and recall, false activations, domain leakage, adaptation delay, context cost, and safety failures. An ablation that activates every preference must not be credited for recall without its precision cost.

10 Human Evaluation

The human-study protocol separates target-user preference judgments from external annotation of correctness and constraints. Participants rate debranded responses before seeing preference explanations. A later transparency task measures whether they can understand, edit, lock, delete, and disable inferred values. Recruitment cannot begin until the applicable institutional review determination is complete.

Not measured in the current study: An approved protocol identifier, recruitment and compensation details, sample flow, preregistered exclusions, rating distributions, reliability before adjudication, blinding assessment, and qualitative coding. No human data have been collected.

11 Discussion

The pilot supports syntactic portability but not yet behavioral conformance: one unchanged structured profile was accepted by two heterogeneous model families, yet the small automatic analysis cannot show that either family produced answers the target user would prefer. That distinction prevents a successful serializer or prompt injector from being mislabeled as successful personalization.

The compiler ablation also exposes a cost-accounting trap. A compressed policy can save input tokens while increasing generated tokens and wall-clock latency. In our sample, directive-like compact clauses appear to have encouraged fuller responses despite a global concise preference. This interpretation is mechanistic and post-hoc; output caps and unequal model sizes are alternative explanations. Future work should jointly optimize adherence, correctness, input tokens, output tokens, and truncation rather than treating prompt length as the efficiency endpoint.

The near-zero compact E-versus-history difference on reference coverage should not be read as equivalence. The interval is wide, the diagnostic is lexical, and only one synthetic persona and four tasks were used. Conversely, the absence of a positive automatic result is useful counterevidence against assuming that typed profiles automatically improve answers. The condition-blinded files now make the next decisive step concrete: target-user preference judgments and independent correctness review under a frozen analysis plan.

The larger research program should distinguish broad benefit, style-only benefit, provider and domain heterogeneity, and net harm from incorrect or stale preferences. Practical effect sizes, correction burden, and abstention quality matter more than statistical significance alone.

Implementation of contextual negative feedback further sharpens the research program. Asking whether a stored preference is true and asking whether it belongs in a current prompt are separable labeling tasks. Future human evaluation should therefore report inference calibration, applicability precision, abstention coverage, and recovery after each error type rather than collapsing them into one personalization score.

12 Privacy, Safety, and Limitations

A portable profile can reduce repeated disclosure but can also centralize sensitive behavioral information. The proposed system is local-first, minimizes raw conversation retention, records provenance, rejects assistant/webpage text as preference authority, and makes profile values user-visible and deletable. Cross-provider use remains a disclosure event and must be explicit and scoped.

Preferences never authorize unsafe output. The representation avoids unnecessary sensitive traits and stores desired explanation behavior rather than demographic inferences. Synthetic privacy canaries are used for leakage tests; real secrets must not be planted.

The benchmark’s controlled personas may exaggerate coherence and stability, automatic checks reward surface compliance, English-only tasks limit cultural and linguistic generalization, provider updates challenge reproducibility, and self-reported familiarity is not a complete expertise measure. Repeated exposure can construct preferences rather than merely reveal them. The history-retrieval

baseline is sensitive to retriever quality. A positive average effect could hide provider-specific or domain-specific harm.

The executed pilot adds several concrete limitations. It uses two quantized local models of very different capacity, one synthetic persona, four English tasks, two replicates, a 260-token cap, and a post-hoc lexical concept diagnostic. Several outputs stopped at the length limit. The local end-point avoids third-party data disclosure but does not represent hosted-provider policies or behavior. Cluster bootstrap intervals over only eight model–task clusters are unstable. The compiler ablation was motivated after inspection of the first run and is therefore hypothesis-generating. No human preference, editability, privacy-leakage, or longitudinal-learning outcome was measured.

Not measured in the current study: Adverse-output, privacy-exposure, subgroup, accessibility, and hosted provider data-retention outcomes. The local pilot contained no participant data.

13 Conclusion

This work frames response personalization as a portable, contextual, transparent, and user-governed policy-selection problem rather than undifferentiated long-term memory. It contributes a typed representation, behavioral-conformance definition, benchmark, executable five-condition harness, local-first product, and blinded human-evaluation path.

Across 128 real generations from two local model families, the same profile and selection pipeline executed without failures. A compact compiler saved 19 input tokens per prompt relative to a short history baseline, but produced about 99 more output tokens and added 3.12 seconds of latency; its exploratory reference coverage was not distinguishable in this small sample. Thus the measured contribution is an auditable systems artifact and a falsifiable efficiency trade-off, not evidence that personalization improves subjective quality. Establishing that claim requires the preregistered target-user study and independent correctness evaluation.

Reproducibility statement

The repository contains versioned benchmark inputs, deterministic mock generation, provider adapters, condition compilation, randomized real local-model runs, raw JSONL responses, immutable input hashes, analysis code, and condition-blinded annotation files. The reported pilot data use Qwen 3 8B and Llama 3.2 1B served locally; all 128 assigned generations completed. No human judgments, private participant data, API credentials, or fabricated outcomes are included.

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